

Rajalakshmi Engineering College

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NeoColab_REC_CS23231_DATA STRUCTURES

REC_DS using C_Week 2_COD_Question 4

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

Ravi is developing a student registration system for a college. To efficiently store and manage the student IDs, he decides to implement a doubly linked list where each node represents a student's ID.

In this system, each student's ID is stored sequentially, and the system needs to display all registered student IDs in the order they were entered.

Implement a program that creates a doubly linked list, inserts student IDs, and displays them in the same order.

Input Format

The first line contains an integer N the number of student IDs.

The second line contains N space-separated integers representing the student IDs.

Output Format

The output should display the single line containing N space-separated integers representing the student IDs stored in the doubly linked list.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5

10 20 30 40 50

Output: 10 20 30 40 50

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
struct node {  
    int id;  
    struct node* prev;  
    struct node* next;  
};
```

```
struct node* head = NULL;
```

```
void append(int id) {  
    struct node* newNode = (struct node*)malloc(sizeof(struct node));  
    newNode->id = id;  
    newNode->prev = NULL;  
    newNode->next = NULL;
```

```
    if (head == NULL) {  
        head = newNode;  
    } else {  
        struct node* temp = head;  
        while (temp->next != NULL) {  
            temp = temp->next;
```

```
    }  
    temp->next = newNode;  
    newNode->prev = temp;  
}  
}
```

```
void display() {  
    struct node* temp = head;  
    while (temp != NULL) {  
        printf("%d ", temp->id);  
        temp = temp->next;  
    }  
    printf("\n");  
}
```

```
int main() {  
    int n, id;  
    scanf("%d", &n);  
  
    for (int i = 0; i < n; i++) {  
        scanf("%d", &id);  
        append(id);  
    }  
  
    display();  
    return 0;  
}
```

Status : Correct

Marks : 10/10